

Seminar 5 (homework)

Discrete probabilities

1. In the poker game there are 52 cards and a player receives 5 cards. Find the probabilities of each of the following hands:

- a. royal flush (ace, king, queen, jack, 10 in same suit);
- b. straight flush (5 cards of 5 consecutive ranks, all in 1 suit, but not a royal flush);
- c. four of a kind (four cards in 1 rank and a fifth card);
- d. full house (3 cards of 1 rank, 2 of another rank);
- e. flush (5 cards in 1 suit, but neither royal nor straight flush);
- f. straight (5 cards in 5 consecutive ranks, but not all of the same suit);
- g. three of a kind (3 cards of 1 rank and 2 cards of 2 different ranks);
- h. two pairs (2 cards in each of 2 different ranks and a fifth card of a third rank);
- i. one pair (2 cards in 1 rank, plus 3 cards from 3 other ranks).

2. (from counting) If $P(n, k)$ denotes the number of integer partitions of n into k parts, then

$$P(n, k) = P(n - k, k) + P(n - k, k - 1) + \dots + P(n - k, 1).$$

Hint: Given a partition of n in k parts, we can obtain a partition of $n - k$ in k (or fewer) parts by subtracting 1 from each part (and conversely, also).